Stanley Nelavala

Software Engineer: Backend, Agent Infrastructure and AI-Native Development

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Huntington, IN | Open to relocate anywhere

Summary

Software engineer who owns features end to end on backend and infrastructure, and builds and scales AI systems: a public LangGraph and MCP agent platform with access labels, human approval, audit log and LLM failover, a gateway for webhooks and connectors, and a Go inference service sized for a 10x peak. I build daily with Claude Code, Codex, opencode and a DeepSeek harness, and run agent fleets under written acceptance checks. M.S. in Computer Science (4.0 GPA). AWS Certified Solutions Architect, Associate.

Experience

Software Development Engineer Nov 2025 – Present
Mass Technologies, team of 10 (RxApps360 hospital platform, pre-launch; numbers from dev and test) *Huntington, IN*

- Designed, built and integrated into every platform module the signed service-to-service API (M2M): each call proves which service and tenant sent it. After a security audit, hardened it with a Redis rate limiter and replay protection.
- Replaced offset paging with keyset paging on the catalog device list: on a 328,000-device dev database, the query for row 150,000 fell from 285 ms to 0.20 ms. Search pages never skip or repeat rows, tested with rows inserted, deleted and tied between pages.
- Fixed a regular expression denial-of-service bug in a shared validation package: an 80-character hostile input took about 138 s; the new linear pattern matched the old results on 60,000 random inputs.
- Built RxDesign, the hospital design service (FastAPI, React, TypeScript; 220+ offline tests). Standardized malware scanning of file uploads (GuardDuty) across 9 app modules.
- Build with Claude Code, Codex and opencode every day: agents write much of the first draft, and I check it with tests, acceptance checks and my own reading before it merges.

Adjunct Faculty (part-time) Jan 2026 – Aug 2026
The College of St. Scholastica *Duluth, MN*

- Taught Programming II (Java) and Machine Learning for Data Science.

Graduate Research Assistant and Teaching Assistant Aug 2023 – Aug 2025
University of Minnesota Duluth *Duluth, MN*

- Built a pipeline (Python, Node.js, MySQL, FFmpeg, PaddleOCR) that collected and analyzed 1M+ TikTok videos from 300k+ authors for thesis research on misinformation, then grouped themes with BERTopic; presented at the UMD CS graduate poster session.

Projects (public code, tests)

Agent Integration Platform | *Python, FastAPI, LangGraph, MCP, PostgreSQL (pgvector), Redis, Terraform* GitHub

- LangGraph case workflow with typed state, a Postgres checkpointer, human approval that resumes after restart, and step and token budgets; one constrained MCP write tool; retrieval with access labels so an agent sees only what it may.
- 62 deterministic replay evals, prompt-injection tests, output guardrails, PII redaction and an append-only audit log; failover between two LLM adapters with timeouts, a retry budget, a circuit breaker and token cost records.

Webhook and Connector Gateway | *Rust, axum, tokio, PostgreSQL, SQLite, sqlx, OpenTelemetry, k6* GitHub

- Rust gateway: HMAC-SHA256 webhooks verified in constant time, per-tenant rate limits and idempotency, durable store before delivery, retries with backoff, dead letters with replay and circuit breakers; 88 tests on Postgres and SQLite.

Inference Gateway | *Go, gRPC, ONNX Runtime, PyTorch, OpenTelemetry, Prometheus, Helm* GitHub

- Go service for an int8-quantized classifier over HTTP and gRPC with micro-batching, deadlines and load shedding; a Little's law model sizes a 10x launch peak at 2 to 3 replicas (local benchmark).

Agent-fleet build workflow | *Claude Code, opencode, DeepSeek, shell* GitHub

- A planner, manager and worker pipeline (Claude plans and manages; opencode and DeepSeek workers build) that delivered two public services and an ops console in 34 work packets, each with written acceptance checks, a progress log and a manager check.

Skills

AI-native development: Claude Code and subagents, Codex, opencode, a DeepSeek worker harness, Claude in Chrome, Gemini, OpenRouter models; planner, manager and worker agent runs with acceptance checks

AI systems: LangGraph, MCP, retrieval (pgvector, RRF), deterministic evals, guardrails, PII redaction, LLM failover, ONNX Runtime, PyTorch

Languages: Python, TypeScript, Go, Rust, Java, C#, SQL

Backend and cloud: FastAPI, axum and tokio, REST, gRPC, PostgreSQL, Redis, Kafka API, AWS (ECS, RDS, S3, SQS), Docker, Terraform, GitHub Actions

Reliability and testing: OpenTelemetry, Prometheus, Grafana, SLOs, k6, pytest, Testcontainers, Playwright

Education and Certification

M.S. Computer Science, University of Minnesota Duluth, 2023 – 2025, GPA 4.0, Graduate Summer Research Fellowship 2024 and 2025

B.Tech Computer Science and Engineering, IIIT D&M Kancheepuram, 2019 – 2023

AWS Certified Solutions Architect, Associate, Feb 2025 – Feb 2028 (verifiable on Credly)